

GISBORNE, New Zealand

WMO: 932920

Lat: 38.6586S

Lon: 177.9850E

Elev: 17

StdP: 14.69

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	36.1	37.8	31.6	25.9	41.1	33.4	28.0	41.8	22.4	56.5	20.1	55.2	4.6	330	0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.3	82.2	67.2	79.7	66.0	77.4	65.1	69.7	77.8	68.2	75.7	66.8	74.1	11.9	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.1	99.6	72.5	65.5	94.2	71.3	64.1	89.6	70.5	33.7	78.0	32.5	76.0	31.4	74.3	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.7	18.2	16.0	DB	32.8	88.5	1.5	3.5	31.7	91.0	30.8	93.1	30.0	95.1	28.8	97.6	(4)
(5)			WB	32.0	72.4	1.7	1.6	30.8	73.6	29.8	74.5	28.8	75.4	27.6	76.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.0	66.6	66.3	63.4	59.1	55.2	50.8	49.9	50.6	53.6	56.8	59.9	64.2
	DBStd	7.47	5.14	4.79	4.35	4.59	4.65	4.34	4.06	4.08	4.45	4.62	5.10	4.89
	HDD50	162	0	0	0	1	7	40	53	42	15	4	0	0
	HDD65	2833	42	37	83	184	306	425	469	446	343	256	167	75
	CDD50	3078	516	456	416	275	168	65	49	61	122	214	296	440
	CDD65	275	93	73	34	8	2	0	0	0	0	2	13	50
	CDH74	1443	569	405	154	18	1	0	0	0	0	8	60	228
	CDH80	203	100	67	10	0	0	0	0	0	0	1	4	21
Wind	WSAvg	6.9	7.3	6.5	6.4	5.8	6.3	6.5	6.6	6.7	7.5	8.1	8.2	7.4
Precipitation	PrecAvg	40.6	2.2	2.8	3.6	3.8	3.5	5.0	4.5	4.2	3.6	2.3	2.3	2.8
	PrecMax	54.6	5.7	7.4	12.7	9.4	11.4	12.9	12.8	11.6	9.4	4.5	4.8	7.0
	PrecMin	29.1	0.4	0.2	0.3	0.5	0.7	1.6	0.9	0.9	0.4	0.6	0.6	0.2
	PrecStd	6.9	1.4	1.9	2.6	2.4	2.4	2.6	2.5	2.6	2.2	1.0	1.1	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.7	86.4	81.0	76.4	71.8	67.0	64.5	65.7	70.8	74.3	79.2	82.8
		MCWB	67.9	69.5	65.7	65.2	62.4	59.2	56.7	56.6	59.5	60.4	63.8	66.8
	2%	DB	82.6	81.7	77.9	72.9	68.5	63.7	61.8	63.0	67.4	71.0	75.4	79.1
		MCWB	67.1	67.9	64.9	62.7	60.0	56.5	54.2	54.3	57.5	59.5	62.4	65.2
	5%	DB	79.8	78.9	75.4	70.1	66.3	61.4	59.8	60.9	64.8	68.5	72.6	76.4
		MCWB	66.1	66.8	64.1	61.4	58.7	55.0	53.1	53.1	55.8	58.1	61.3	64.4
	10%	DB	77.1	75.9	72.6	67.8	64.0	59.5	57.7	58.9	62.5	66.3	70.0	74.0
		MCWB	65.0	65.8	63.5	60.9	57.4	53.6	52.0	52.2	54.6	57.0	59.6	63.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.3	72.6	69.2	67.2	64.4	60.9	58.3	58.4	60.8	62.8	66.2	69.1
		MCDB	81.5	80.0	76.1	72.5	68.4	64.9	62.5	63.3	68.5	71.6	75.9	78.5
	2%	WB	69.4	70.1	67.1	65.0	62.2	58.2	55.8	56.0	58.6	60.7	63.9	67.1
		MCDB	78.3	77.5	73.1	69.6	66.7	61.9	59.8	60.3	65.0	68.8	72.6	75.7
	5%	WB	67.8	68.5	65.7	63.3	60.1	56.3	54.3	54.3	57.2	59.2	62.3	65.6
		MCDB	76.0	75.7	72.0	67.9	64.5	60.1	58.1	58.7	63.1	66.5	70.6	73.7
	10%	WB	66.4	67.0	64.4	61.8	58.2	54.3	52.8	53.0	55.5	57.8	60.7	64.3
		MCDB	74.1	74.0	71.0	66.4	62.8	58.6	56.8	57.6	61.1	64.7	68.7	71.9
Mean Daily Temperature Range	5% DB	MDBR	16.3	15.6	15.6	15.6	15.3	14.8	13.7	14.9	15.7	16.0	15.9	15.4
		MCDBR	21.8	20.6	20.7	19.0	17.7	16.8	16.0	17.7	19.1	19.3	20.2	19.9
		MCWBR	9.7	9.7	10.5	10.7	10.8	11.2	10.4	11.0	11.5	11.0	10.9	10.0
	5% WB	MCDBR	18.0	16.7	16.5	14.3	14.1	13.9	12.9	14.9	16.7	17.0	18.1	16.8
		MCWBR	9.1	8.9	9.5	8.9	9.6	10.3	9.4	10.3	10.8	10.5	10.7	9.3
Clear-Sky Solar Irradiance	taub	0.348	0.343	0.330	0.314	0.301	0.295	0.298	0.300	0.322	0.334	0.335	0.347	
	taud	2.539	2.575	2.606	2.639	2.647	2.634	2.627	2.627	2.537	2.514	2.531	2.538	
	Ebn at Noon	312	306	298	283	268	259	266	284	295	305	314	314	
	Edh at Noon	34	32	28	24	21	19	21	24	30	33	34	35	
All-Sky Solar Radiation	RadAvg	2116	1784	1415	1011	720	564	615	888	1250	1678	1939	2053	
	RadStd	131	128	116	73	52	37	46	57	119	95	125	147	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.45	+0.63	+0.77	N/A	N/A	+0.59	-32	-137	+155	N/A
(47) Regional (4 neighbors)		+0.52	+0.68	+1.03	+0.83	N/A	N/A	-31	-151	+155	N/A

Nomenclature: See separate page